

Diagnostic — Week 1 (GCSE Bridging)

Name: _____ Date: _____ Mark: _____ / 40

OCR H446 · Week 1 · 45 minutes · Closed book. This paper checks the GCSE foundations the A Level builds on — nothing beyond GCSE Computer Science is assumed. Show your working on every calculation.

Section A — Number bases and data (12 marks)

1. Convert the 8-bit binary number 10110101 to denary. [1]

2. Convert the denary number 156 to 8-bit binary. [1]

3. Convert the denary number 202 to hexadecimal. Show your working. [2]

4. Convert the hexadecimal number 4F to denary. [1]

5. Add the 8-bit binary numbers 11010110 and 01001011. Give your answer in binary, and state whether an overflow error occurs in an 8-bit register, giving a reason. [3]

6. The binary number 00011011 is shifted **two places to the left**. Write the result and state the effect the shift has on the value. [2]

7. An image is 100 pixels wide and 50 pixels high, with a colour depth of 8 bits. Calculate the file size of the image in **bytes**. Show your working. [2]

Section B — Computational thinking and reading pseudocode (14 marks)

8. State what is meant by **decomposition** and what is meant by **abstraction**. [2]

9. The following program is run. [4]

```
total = 0
for i = 1 to 4
    total = total + i * i
next i
print(total)
```

Complete the trace table, adding one row for each time round the loop, and state the output.

i	total	Output

10. The following program is run three times. [3]

```

mark = int(input("Enter the mark"))
if mark >= 70 then
    print("Distinction")
elseif mark >= 40 then
    print("Pass")
else
    print("Fail")
endif

```

Complete the table to show the output for each value entered.

Value entered	Output
70	
39	
55	

11. The following program is meant to count down from 10 to 1, but it does not work. Identify the error and write the line of code that fixes it, stating where it goes. **[2]**

```

count = 10
while count > 0
    print(count)
endwhile

```

12. The following program is run.

```

values = [12, 5, 19, 8]
biggest = values[0]
for i = 1 to values.length - 1
    if values[i] > biggest then
        biggest = values[i]
    endif
next i
print(biggest)

```

(a) State the output of the program. **[1]**

(b) State the purpose of the algorithm. [1]

(c) State the change needed so that the program finds the **smallest** value instead. [1]

Section C — Computer systems basics (14 marks)

13. State the purpose of the **CPU** and name **two** components found inside it. [3]

14. Give **two** differences between **RAM** and **ROM**. [2]

15. A student needs to carry their coursework files between school and home. State why a computer needs **secondary storage**, recommend a suitable storage device for the student, and justify your choice. [3]

16. State **two** functions of an **operating system**. [2]

17. The computers in a school are connected in a network. State **one** advantage and **one** disadvantage of connecting computers in a network. **[2]**

18. State what is meant by an **embedded system** and give **one** example. **[2]**
